```
global_data = {

    # ── Attribute Names
    "attribute_1": "total_cost",                # USD per 1 km unit
    "attribute_2": "material_continuity",       # 0.0–1.0 scale
    "attribute_3": "electrical_power_losses",   # kW/km at rated current
    "attribute_4": "cable_weight",              # kg/m
    "attribute_5": "cable_outer_diameter",      # mm

    # ── Optimization Directions
    "direction_attribute_1": "negative",   # lower cost is better
    "direction_attribute_2": "positive",   # higher material continuity is better
    "direction_attribute_3": "negative",   # lower losses are better
    "direction_attribute_4": "negative",   # lighter cable is better
    "direction_attribute_5": "negative",   # smaller diameter is better

    # ── Global Min / Max for Normalization

    # total_cost: spans from a bare-minimum Al cable in a low-cost greenfield
    # market all the way to a premium Cu floating-wind dynamic cable with
    # top-tier North Sea/US installation vessel rates
    "attribute_1_min": 10000,    # USD/km
    "attribute_1_max": 400000,   # USD/km

    # material_continuity: definitionally 0–1; no physical value exists outside
    "attribute_2_min": 0.0,
    "attribute_2_max": 1.0,

    # electrical_power_losses: lower end = oversized 630mm² Cu cable at low load;
    # upper end = small/undersized cable at peak load or high-resistance Al
    # in hot burial environment
    "attribute_3_min": 10,       # kW/km
    "attribute_3_max": 120,      # kW/km

    # cable_weight: lower end = lightweight dynamic riser, minimal armour;
    # upper end = triple-armoured deep-water or rock-scour-protected cable
    "attribute_4_min": 8,        # kg/m
    "attribute_4_max": 120,      # kg/m

    # cable_outer_diameter: lower end = compact single-core or HVDC feeder;
    # upper end = large triple-armoured 66kV heavy-wall sheathed cable
    "attribute_5_min": 50,       # mm
    "attribute_5_max": 250,      # mm
```

```
    # ── Commodity Prices
    "copper_price_per_ton": 9500,
    "aluminum_price_per_ton": 2500,
    "copper_price_per_kg": 9.5,
    "aluminum_price_per_kg": 2.5,
}


regional_data = [
    {
        "region": "United States",
        # Offshore wind nascent (~2020s ramp); Jones Act cost pressure; gr
        "copper_product_market_share":   0.80,   # Cu still dominant in new US offshore build-out
        "aluminum_product_market_share": 0.20,   # Al gaining ground in cost-comp
        "weight_attribute_1": 0.30,   # total_cost          — high, but PPA struct
        "weight_attribute_2": 0.15,   # material_continuity — moderate; US lack
        "weight_attribute_3": 0.25,   # power_losses        — significant; high electri
        "weight_attribute_4": 0.15,   # cable_weight        — relevant; Pacific/Gulf
        "weight_attribute_5": 0.15,   # outer_diameter      — relevant; J-tubes bein
        # SUM = 1.00 ✓
    },
    {
        "region": "China",
        # World's largest offshore wind deployer; state-driven cost competit
        "copper_product_market_share":   0.75,
        "aluminum_product_market_share": 0.25,

        "weight_attribute_1": 0.40,   # total_cost          — dominant; competitive govt tend
        "weight_attribute_2": 0.20,   # material_continuity — strong Cu SOE supply cha
        "weight_attribute_3": 0.20,   # power_losses        — matters at China's deployment
        "weight_attribute_4": 0.10,   # cable_weight        — low; shallow fixed-bottom, no floa
        "weight_attribute_5": 0.10,   # outer_diameter      — low; flexible domestic manufact
        # SUM = 1.00 ✓
    },
    {
        "region": "India",
        # Greenfield offshore market; extreme cost sensitivity; SECI/MNRE budget-driven tenders
        "copper_product_market_share":   0.65,
        "aluminum_product_market_share": 0.35,   # highest Al share — cost pressure driv
        "weight_attribute_1": 0.45,   # total_cost          — overwhelmingly dominant; gre
        "weight_attribute_2": 0.10,   # material_continuity — lowest weight; no establishe
        "weight_attribute_3": 0.25,   # power_losses        — meaningful; 25-yr asset
        "weight_attribute_4": 0.10,   # cable_weight        — low; Gujarat/TN fixed
        "weight_attribute_5": 0.10,   # outer_diameter      — low; greenfield projects all
```

```
        # SUM = 1.00 ✓
    },
    {
        "region": "Europe",
        # Most mature offshore wind market (30+ yrs); lifecycle-optimised procureme
        "copper_product_market_share":   0.75,
        "aluminum_product_market_share": 0.25,

        "weight_attribute_1": 0.25,   # total_cost          — important but balanced by LCO
        "weight_attribute_2": 0.20,   # material_continuity — high; deep institutional Cu m
        "weight_attribute_3": 0.30,   # power_losses        — HIGHEST weight; sophisticated
        "weight_attribute_4": 0.15,   # cable_weight        — relevant; ScotWind/Norway floati
        "weight_attribute_5": 0.10,   # outer_diameter      — moderate; legacy North Se
        # SUM = 1.00 ✓
    },
]


product_data = [


    {
        "region":           "United States",
        "dominant material": "cu",
        "product":       "33kV 3-Core 185mm² Cu XLPE SWA Submarine Inter-Array Cable (1km)",

        "copper_kg":              5200,    # 3×185mm² conductors + Cu tape screens (~5,000 +
        "aluminum_kg":               0,    # no aluminum content
        "total_mass_kg":         40000,    # 40 kg/m × 1000 m (conductor + XLPE + SWA + HDPE)

        "non_material_cost_per_unit": 85000,
        # Jones Act-compliant vessels, US harbor labour, XLPE extrusion, SWA armouring
        # total_cost = (5200 × 9.5) + (0 × 2.5) + 85,000 = $134,400 /km

        "attribute_2_value": 0.70,  # material_continuity — moderate; US offshore Cu hist
        "attribute_3_value": 47,    # kW/km losses  — Cu 185mm² at 350A, 3-phase, 90°C
        "attribute_4_value": 40,    # kg/m
        "attribute_5_value": 118,   # mm outer diameter
    },
    {
        "region":           "United States",
        "dominant material": "al",
        "product":          "33kV 3-Core 300mm² Al XLPE SWA Submarine Inter-Array Cable (1km)",
```

```
    "copper_kg":              400,    # Cu tape screens + earth continuity conductor
    "aluminum_kg":            2430,    # 3×300mm² Al conductors (300e-6 m² ×
    "total_mass_kg":         34000,    # 34 kg/m × 1000m (lighter conductor, similar armour)

    "non_material_cost_per_unit": 78000,
    # Slightly lower than Cu (simpler conductor draw), same high US installation overhead
    # total_cost = (400 × 9.5) + (2430 × 2.5) + 78,000 = $87,875 /km

    "attribute_2_value": 0.30,  # material_continuity — low; US market built on Cu, A
    "attribute_3_value": 52,    # kW/km — marginally higher due to larger buried cable t
    "attribute_4_value": 34,    # kg/m  — 15% lighter than Cu equivalent
    "attribute_5_value": 133,   # mm    — ~13% larger diameter than Cu equivalent
  },
  {
    "region":          "China",
    "dominant material": "cu",
    "product":     "33kV 3-Core 185mm² Cu XLPE SWA Submarine Inter-Array Cable (1km)",

    "copper_kg":             5200,
    "aluminum_kg":              0,
    "total_mass_kg":        40000,

    "non_material_cost_per_unit": 45000,
    # Lowest non-material cost: domestic Hengtong/ZTT manufacturing, local lab
    # total_cost = (5200 × 9.5) + (0 × 2.5) + 45,000 = $94,400 /km

    "attribute_2_value": 0.80,  # material_continuity — strong; SOE Cu supply ch
    "attribute_3_value": 47,
    "attribute_4_value": 40,
    "attribute_5_value": 118,
  },
  {
    "region":          "China",
    "dominant material": "al",
    "product":        "33kV 3-Core 300mm² Al XLPE SWA Submarine Inter-Array Cable (1km)",

    "copper_kg":              400,
    "aluminum_kg":           2430,
    "total_mass_kg":        34000,

    "non_material_cost_per_unit": 40000,
    # Cheapest Al non-material cost globally: fully domestic supply chain
    # total_cost = (400 × 9.5) + (2430 × 2.5) + 40,000 = $49,875 /km
```

```
    "attribute_2_value": 0.20,  # material_continuity — low; offshore wind in Chi
    "attribute_3_value": 52,
    "attribute_4_value": 34,
    "attribute_5_value": 133,
},

{
    "region":           "India",
    "dominant material": "cu",
    "product":      "33kV 3-Core 185mm² Cu XLPE SWA Submarine Inter-Array Cable (1km)",

    "copper_kg":              5200,
    "aluminum_kg":               0,
    "total_mass_kg":         40000,

    "non_material_cost_per_unit": 38000,
    # Low labour cost advantage; most cable imported or assembled locally from components
    # total_cost = (5200 × 9.5) + (0 × 2.5) + 38,000 = $87,400 /km

    "attribute_2_value": 0.50,  # material_continuity — neutral; no established offshore
    "attribute_3_value": 47,
    "attribute_4_value": 40,
    "attribute_5_value": 118,
},
{
    "region":           "India",
    "dominant material": "al",
    "product":          "33kV 3-Core 300mm² Al XLPE SWA Submarine Inter-Array Cable (1km)",

    "copper_kg":              400,
    "aluminum_kg":            2430,
    "total_mass_kg":         34000,

    "non_material_cost_per_unit": 34000,
    # Lowest non-material cost globally; cost-optimised greenfield procurement
    # total_cost = (400 × 9.5) + (2430 × 2.5) + 34,000 = $43,875 /km

    "attribute_2_value": 0.50,  # material_continuity — neutral; no legacy pref
    "attribute_3_value": 52,
    "attribute_4_value": 34,
    "attribute_5_value": 133,
},

{
```

```
    "region":          "Europe",
    "dominant material": "cu",
    "product":      "33kV 3-Core 185mm² Cu XLPE SWA Submarine Inter-Array Cable (1km)",

    "copper_kg":              5200,
    "aluminum_kg":               0,
    "total_mass_kg":         40000,

    "non_material_cost_per_unit": 78000,
    # High-quality Prysmian/NKT/Nexans production; EU labour rates;
    # total_cost = (5200 × 9.5) + (0 × 2.5) + 78,000 = $127,400 /km

    "attribute_2_value": 0.90,  # material_continuity — very strong; 30+ yrs of Cu off
    "attribute_3_value": 47,
    "attribute_4_value": 40,
    "attribute_5_value": 118,
  },
  {
    "region":          "Europe",
    "dominant material": "al",
    "product":         "33kV 3-Core 300mm² Al XLPE SWA Submarine Inter-Array Cable (1km)",

    "copper_kg":              400,
    "aluminum_kg":            2430,
    "total_mass_kg":         34000,

    "non_material_cost_per_unit": 72000,
    # Lower conductor cost offset by same high EU manufacturing and installation overhead
    # total_cost = (400 × 9.5) + (2430 × 2.5) + 72,000 = $81,875 /km

    "attribute_2_value": 0.10,  # material_continuity — very low; Europe has
    "attribute_3_value": 52,
    "attribute_4_value": 34,
    "attribute_5_value": 133,
  },
]
```